Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	**Early tumor detection using high-sensitivity, broadband endocavity ultrasound probe for volumetric imaging at high spatial resolution**			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (√), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1 (**Large Category**)**		**Technology Field 2 (**Large Category**)**	
	Bio · Medical		Mechanical · Materials	
	Medium Category	Small Category	Medium Category	Small Category
	Therapeutic Devices & Diagnostic Devices	Ultrasound Diagnostic Devices	Robotics/Automation Machinery	Mechanical Automation Technologies
Concept of Proposed Technology	■ **Background and Problem Definition** · Overdiagnosis and false positives remain major clinical challenges in cancer prevention and early detection. There is an unmet medical need to improve the accuracy of identifying aggressive early-stage tumors and differentiating clinically significant lesions from non-life-threatening abnormalities. · Volumetric endocavity ultrasound (US) and photoacoustic (PA) imaging can provide close-proximity, high-resolution visualization of potential cancer regions. However, current rod-shaped probes based on conventional one-dimensional piezoelectric arrays exhibit fundamental technical limitations: 1. **Insufficient Acoustic Bandwidth and Sensitivity**[1-7]**:** Suboptimal acoustic stack designs limit bandwidth and energy transmission, reducing sensitivity to endogenous and exogenous tumor contrast signals at clinically relevant depths. 2. **Limited Spatial Resolution:** Elevational resolution is restricted by acoustic lens–based focusing, which degrades beam quality outside the depth of focus. 3. **Geometric Constraints:** Endocavity entry size restricts probe geometry and prevents implementation of advanced			

	volumetric imaging techniques such as synthetic aperture focusing[8-11]. ■ **Core Concept** • This project proposes a **novel deployable endocavity ultrasound and photoacoustic probe architecture** that overcomes the above limitations by integrating synthetic aperture compatible, and high-performance acoustic module design with analytically validated volumetric imaging principles. • The ultrasound probe is redefined as **mechatronic-reconfigurable active imaging component**, enabling customizable acoustic beam profiles[12-14] optimized for volumetric imaging. A recent development of a radial synthetic aperture focusing (rSAF) strategy[8-9] opens an opportunity for endocavity US and PA imaging to significantly enhance radial spatial resolution while preserving clinical usability under endocavity constraints by expanding the effective scanning radius only during acquisition. • The core objective is to realize a deployable probe that simultaneously achieves: 1. Synthetic aperture compatible, high-performance acoustic module design 2. Structural compatibility with rSAF-based volumetric imaging ■ **Technical Approach** • The probe maintains a conventional external form factor during insertion and transitions to an imaging mode after positioning at the region of interest. In imaging mode, a miniaturized internal mechanism expands the effective scanning radius while a synthetic aperture compatible, high-performance acoustic module ensures efficient acoustic energy transmission and broad bandwidth to provide sufficient signal-to-noise ratio for synthetic aperture-based volumetric focusing. ✓ The Korean SME partner leads probe-level design, including synthetic aperture compatible, high-performance acoustic module optimization, deployable geometry design, fabrication, and characterization. ✓ The research institute provides analytical rSAF frameworks, defines target imaging configurations, develops reconstruction algorithms, and supports phantom-based validation. ■ **Main Functions** • High-performance broadband ultrasound transmission and reception characteristics[12-14] • Miniaturized mechatronics for expandable scanning trajectories[8-11,15] • Rapid volumetric image reconstruction software • Compatibility with conventional ultrasound imaging platforms ■ **Clinical Impact** • The high-quality volumetric imaging enabled by this technology simultaneously improves sensitivity and specificity for early-stage tumor detection in endocavity imaging[16]. Enhanced spatial resolution strengthens the ability to identify clinically significant small lesions while structurally reducing dependence on operator-dependent interpretation. Furthermore, in interventional settings, the proposed approach enables stable and consistent volumetric image guidance without altering existing clinical workflows, thereby improving procedural accuracy and reproducibility.
Rationale for Joint R&D Project	• Advanced volumetric endocavity imaging performance is constrained primarily by probe-level acoustic transmission rather than imaging algorithms alone. While synthetic aperture focusing frameworks analytically assume broadband, high-sensitivity, and stable acoustic behavior, these conditions cannot be physically realized under endocavity constraints without probe architectures explicitly designed to satisfy imaging-level assumptions. • Joint R&D is required to integrate analytically defined volumetric imaging requirements with physically realizable, mechatronically deployable probe architectures through coordinated development and experimental validation. Without such collaboration, volumetric imaging concepts remain theoretically sound but practically unrealizable, while probe development is confined to component-level optimization without system-level imaging objectives. • This joint framework ensures the end-to-end translation of advanced volumetric imaging theory

	into deployable endocavity probe systems, while simultaneously strengthening domestic SME capabilities in system-level probe architecture ownership beyond conventional component manufacturing.
Key Words	• Endocavity ultrasound and photoacoustic imaging • Radial synthetic aperture focusing (rSAF) for Volumetric ultrasound imaging • Advanced acoustic matching materials and architectures • Deployable probe architecture • Mechatonics-integrated probe

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL) [2]	**Start of the Project**	Level (2)	**End of the Project**	Level (6)
R&D Objectives	1. Realize a Clinically Usable rSAF-based Endocavity Deployable Probe Architecture ✓ Realize a mechanically deployable probe enabling controlled aperture expansion under strict endocavity clinical constraints. 2. Achieve rSAF-compatible Acoustic Performance enabled by Advanced Material Engineering ✓ Establish deployment-invariant acoustic performance through advanced acoustic stack and material engineering, ensuring stable synthetic aperture imaging beyond static operating conditions. 3. Experimental Validation of Volumetric Imaging Performance ✓ Quantitatively validate rSAF imaging performance, including resolution, SNR, grating lobe behavior, and deployment-state-dependent stability using phantom studies 4. Internalization of Probe-Imaging Co-Design Capability ✓ Internalize integrated acoustic, mechanical, and imaging co-design workflows within the Korean SME and assess regulatory feasibility toward clinical translation. 5. Establish a Foundation for Commercialization and Regulatory Transition			
R&D Target Specifications	■ **Deployment-Invariant Acoustic Performance** 1. Dynamic control of deployable imaging radius ≥ 2.5 × insertion-mode (e.g., expansion from 2 cm to 5 cm) at array location encoding precision less than half lambda. 2. Fractional bandwidth ≥ 70% with impulse response ≤ 2.5 cycles 3. Deployment repeatability (CV) ≤ 10% ■ **rSAF-Based Volumetric Imaging Performance** 1. Measured rSAF resolution within ±15% of analytical predictions 2. No increase in electronic channel count			
R&D Plan	■ **Phase 1 (TRL 2 → 4): Definition of Acoustic Requirements and Probe Architecture** • Derive rSAF acoustic requirements • Design rSAF-compatible acoustic stack architecture • Characterize baseline and optimized probe performance • Conceptual design deployable probe structures ■ **Phase 2 (TRL 4 → 5): Deployable Probe Integration and Imaging Feasibility** • Detailed design and fabricate deployable probe structures • Verify and validate acoustic stability before and after deployment, and repeatability • Demonstrate rSAF-based imaging feasibility ■ **Phase 3 (TRL 5 → 6): Integrated rSAF Validation** • Conduct volumetric imaging experiments • Quantitatively evaluate volumetric imaging performance including spatial resolution, SNR,			

	grating lobe behavior • Establish validated design and operating boundaries

4. R&R (Roles and Responsibilities)

Expected R&R of Research Instituter	1. Provide rSAF analytical framework and quantitative design specifications 2. Develop reconstruction algorithms and imaging performance metrics 3. Design validation protocols and interpret analytical-experimental consistency 4. Perform joint experimental validation using an integrated probe with imaging systems 5. Lead technical reviews
Expected R&R of Korean SME Partner	1. Lead deployable probe architecture design and fabrication 2. Develop performance-optimized and deployment-robust acoustic structures 3. Lead Integrate probe with imaging systems and perform experimental validation 4. Establish manufacturability, repeatability, and quality-control workflows 5. Internalize advanced probe design and system integration capabilities

5. Expected Outcomes/Impact

Expected Outcomes and Impact	■ **Expected Outcomes** • Prototype deployable rSAF-based volumetric endocavity US and PA probe • Experimental validation of rSAF analytical imaging predictions • Established design envelope for deployable rSAF-compatible probe architectures • Internalized probe-imaging co-design workflows of the Korean SME • Growth of a Korean SME into a multidisciplinary probe developer integrating mechatronics and acoustic probe design • Foundation for commercialization and regulatory transition ■ **Expected Impact** • System-level realization and validation of deployable rSAF-based volumetric endocavity imaging • Experimentally verified linkage between deployable probe acoustics and rSAF imaging validity • Probe architecture ownership and co-design capability established within the Korean SME • Scalable foundation for next-generation volumetric ultrasound probe development • Stable and interpretable volumetric endocavity imaging without workflow modification • Reference model for international joint R&D translating imaging theory into deployable medical probe systems • Creation of new-market revenue opportunities for the Korean SME through commercialization of deployable rSAF-based probe technologies

6. References

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